

Daeyang University

College of ICT

Name of Programme: Information and Communication Technology

Full Title of Assignment (s)

Organization's Name: Smart Applications International

Industrial Attachment Supervisor: Mr Joshua Muteti

Student's Name: Dennis Wella

ID: BScICT/19/132

Date of submission: January 14, 2024

CERTIFICATE OF ORIGINALITY

I hereby declare that the intellectual content of this portfolio is the product of my own work and that, to the best of my knowledge and belief, it contains no material previously published or written by another person nor material which has been accepted for the award of any other degree or diploma of the University or any other institute, except where due acknowledgement and references are made in the text.

Candidate's Name/Number:

Assignment Title:

Signature:

Date:

Acknowledgements

I would like to extend my heartfelt gratitude to the faculty and staff at Daeyang University's College of ICT for providing me with the academic foundation and skills necessary for a successful industrial attachment. Special appreciation goes to my Industrial Attachment Supervisor, [Supervisor's Name], whose guidance and mentorship were invaluable throughout this experience. I am also thankful for the support from my colleagues at [Organization's Name], whose collaboration made the workplace a conducive learning environment. Lastly, I express my sincere thanks to my family and friends for their unwavering encouragement and understanding during this challenging yet rewarding journey.

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Introduction.

Background

In the dynamic landscape of Information and Communication Technology (ICT), the synergy between academic learning and practical experience is crucial for the comprehensive development of students. This portfolio chronicles my transformative expedition during the Final Year Industrial Attachments as part of the ICT program at Daeyang University's College of ICT. The amalgamation of theoretical knowledge with hands-on application in a real-world environment has been a cornerstone of my educational journey.

Objectives

The primary objective of this industrial attachment was to bridge the gap between classroom learning and industry demands. This immersion aimed to provide insights into the practical aspects of ICT, fostering a deeper understanding of the field and enhancing my adaptability to professional challenges.

Scope

The scope of this portfolio encompasses a comprehensive exploration of my industrial attachment at Smart application international. It delves into the daily responsibilities, challenges encountered, and the application of theoretical knowledge in real-world scenarios.

Significance of the Study

This portfolio not only serves as a reflective documentation of my individual growth but also contributes to the collective knowledge base of Daeyang University's academic community. By sharing my experiences and insights, I hope to inspire future students to embrace the opportunities presented by industrial attachments, fostering a culture of experiential learning within the College of ICT. As the technological landscape continues to advance, the significance of hands-on experiences becomes increasingly vital for producing well-rounded and industry-ready professionals.

Brief Overview of the Organization.

Smart Applications International is a leading ICT solutions provider delivering a wide range of world-class technological solutions. Fondly known as Smart within the industry, the company was founded to provide solutions that transform organizational processes to achieve effective identity management, automation, operational efficiency, and cost savings. We have partnered with medical service providers in all 47 counties at over 5000 hospitals in Kenya to ensure system availability in the access to medical services. This has as well been extended to 400 providers in Uganda, 300 providers in Rwanda, 200 providers in Zambia, 100 providers in South Sudan, 100 providers in Somalia, and 29 providers in Tanzania with a regional tally of 10 countries in Africa.

Location and History

Founded in 2004 Smart Applications International Ltd is based in Nairobi, Nairobi County with 248 people are employed and an annual revenue of \$6 million in 2023

Mission

Inspiring a world of convenience

Vision

Create a Smarter Society Through World Class Innovative Technology Solutions

Core Values

Our People: Are our heroes

Integrity: Our word is our bond

Innovation: Innovation is our end game

Customers: We are for you, by you, with you

Security: Peace of mind

Services and Programs

Smart Applications, a leading ICT provider of biometric solutions, offers an extensive range of innovative solutions, including the Automated Medical Scheme Management Solution. Our dedicated team of software developers creates simple and user-friendly applications to address day-to-day business challenges. We provide expert advice to clients on the most effective and efficient systems for their business needs, and our service and support desk is

accessible 24/7/365 to address any inquiries promptly. At Smart Applications, we empower businesses to make informed decisions by providing meaningful interpretations of data through cutting-edge technology and analytics tools, delivering valuable business intelligence.

Experience Gained, Activities Covered, Application of theoretical knowledge in practice

Daily Tasks and Responsibilities

During my internship, I served as a 24/7 technical support resource for both the smartlink system and the smarthealth hospital information management system, addressing and resolving technical issues to maintain optimal functionality. I conducted on-site visits to hospitals, performed necessary system updates, proactively identified hospitals without a hospital information management system, pitched the benefits of the smarthealth solution, and followed up diligently. Additionally, I actively contributed to the implementation of the smarthealth system, installing the smartlink system at hospitals when instructed to do so, and played a crucial role in the ongoing maintenance tasks, ensuring overall system efficiency and reliability.

technical support

During my internship, I assumed the critical role of being a 24/7 technical support resource for both the smartlink system and the smarthealth hospital information management system, providing assistance and troubleshooting over the phone. This responsibility required me to address and resolve a myriad of technical issues promptly and effectively, ensuring uninterrupted system functionality for healthcare providers. Whether it was diagnosing and resolving issues remotely or guiding users through step-by-step solutions over the phone.

on-site visits

I conducted on-site visits to various hospitals, playing a crucial role in ensuring the optimal performance of both the smartlink system and the smarthealth hospital information management system. During these visits, I meticulously assessed the operational status of the systems, identifying any potential issues or areas for improvement. This hands-on approach allowed me to interact directly with healthcare professionals, gaining valuable insights into their user experience and understanding the unique challenges they faced. In addition to these visits, I took charge of performing necessary system updates to guarantee that the technology remained current and aligned with the latest industry standards. This involved implementing software upgrades and system enhancements to enhance functionality and address any vulnerabilities. By staying proactive in these updates, I contributed to the overall reliability and efficiency of the smartlink and smarthealth systems, ensuring that healthcare providers could rely on these technologies for seamless operations in their day-to-day activities. Below is a sample of the itinerary I would use to plan my on-site visits:

DAY	PROVIDER_KEY	PROVIDER NAME	OFFICER IN CHARGE	Purpose for Visit
TUESDAY(25-JULY)	SMWSP_132	CMED	Dennis	.Deploying SBB .Training on SBB
	SMWSP_6	GOOD HOPE PRIVATE CLINIC	Dennis	.Deploying SBB .Training on SBB
	SMWSP_95	NEW HORIZON HEALTHCARE	Dennis	.Deploying SBB .Training on SBB
WEDNESDAY(26-JULY)	SMWSP_10	LILONGWE ADVENTIST HOSPITAL	dennis	.Deploying SBB .Training on SBB
	SMWSP_7	CITY CENTRE CLINIC	dennis	.Deploying SBB .Training on SBB
	SMWSP_123	FAMILY DENTAL CLINIC	dennis	.Deploying SBB .Training on SBB
THURSDAY(27-JULY)	SMWSP_9	LILONGWE PRIVATE CLINIC	dennis	.Deploying SBB .Training on SBB
	SMWSP_31	WEM DENTAL PRIVATE CLINIC	dennis	.Deploying SBB .Training on SBB
	SMWSP_12	MASM MEDICLINIC- AREA 12	dennis	.Deploying SBB .Training on SBB
FRIDAY(28-JULY)	SMWSP_5	LIKUNI MISSION HOSPITAL	dennis	.Deploying SBB .Training on SBB
	SMWSP_4	TACHIRA PRIVATE CLINIC	dennis	.Deploying SBB .Training on SBB
	SMWSP_3	THE CLINIC	dennis	.Deploying SBB .Training on SBB
	SMWSP_32	PHARMACARE PHARMACIES	dennis	.Deploying SBB .Training on SBB
	SMWSP_50	FALLS BANJA LA MTSOGOLO	dennis	.Deploying SBB .Training on SBB
MONDAY(31-JULY)	SMWSP_94	SIGHTCARE OPTICALS	dennis	.Deploying SBB .Training on SBB
	SMWSP_11	MASM MEDICLINIC-LINGADZI	dennis	.Deploying SBB .Training on SBB
	SMWSP_121	DISCOVERY IMAGING CENTRE	dennis	.Deploying SBB .Training on SBB
	SMWSP_130	SSH CLINIC	dennis	.Deploying SBB .Training on SBB

Figure 1.1

Smart health

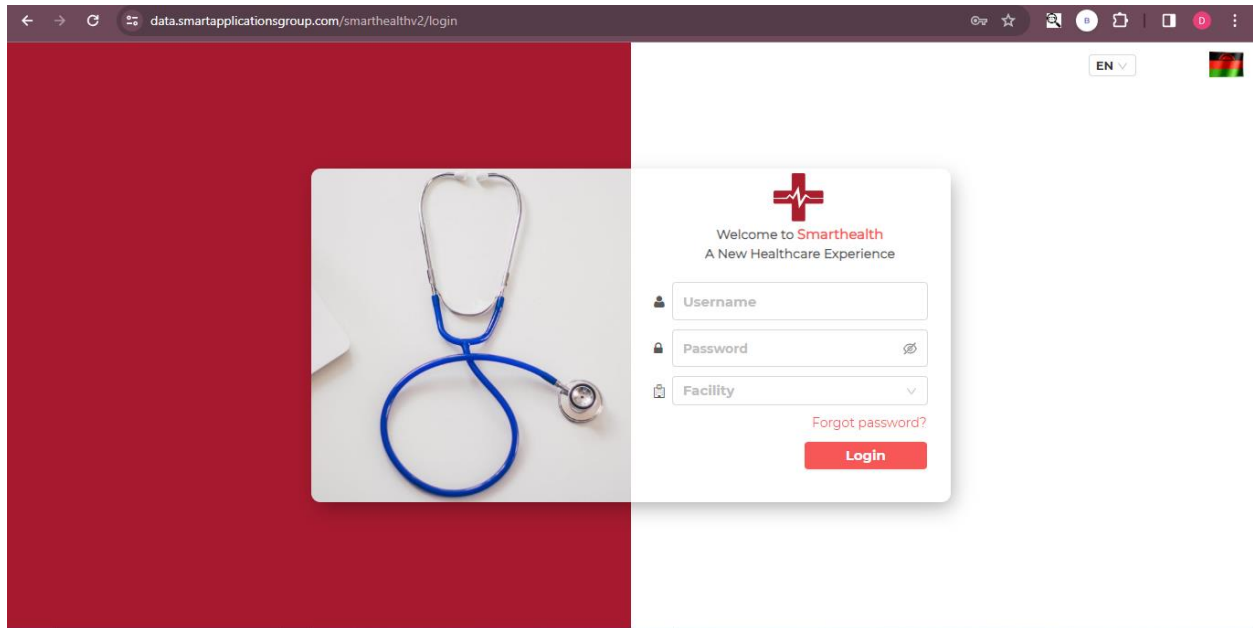
In a proactive capacity during my internship, I took the initiative to identify healthcare facilities lacking a hospital information management system (HIMS). Once identified, I enthusiastically embarked on the role of a solution advocate. Through persuasive presentations and discussions, I conveyed how adopting smarthealth could streamline their administrative processes, enhance data accuracy, and ultimately improve overall patient care.

implementation of the smarthealth

I played a pivotal role in the active contribution to the implementation phase of the smarthealth hospital information management system. This multifaceted responsibility involved various key tasks to ensure a seamless integration and effective utilization of the system within the healthcare facilities. I took charge of conducting comprehensive training sessions for all users of the system. This included healthcare professionals, administrative staff, and any other individuals who would interact with the smarthealth system. Through these training sessions, I ensured that users had a comprehensive understanding of the system's functionalities, fostering a smooth transition to its utilization in their day-to-day operations. I undertook the technical aspects of the implementation process. This included setting up a local area network (LAN), I played a crucial role in entering essential information into the system. This involved meticulously inputting data such as prices for medicine, procedures, and lab tests for both private and public sections of the hospitals. I also recorded critical infrastructure details, including the number of beds in the ward and freezers in the morgue. I actively contributed to the establishment of user accounts by creating profiles for individuals who would be interacting with the system. This extended to assigning roles and rights to users, ensuring that each individual had the appropriate level of access aligned with their responsibilities within the healthcare facility.

Smarthealth

In the "Smarthealth" system, everyone gets their own username and password to log in. When logging in for the first time, you are prompted to create a new password, which must have 8 characters, a capital letter, a number, and a symbol for security reasons. As shown in Figure 2.1, the user will enter their login details, select the facility they are operating in from the drop-down menu, and then press "login."



Once logged in, you are greeted with a dashboard full of information about your facility. On the left side of the page, as seen in Figure 2.2, you can see the list of modules that a username can access. In the "Smarthealth" system, not all users can access all modules. Only users logged in as administrators can view all the modules in the system. For example, a receptionist will only have access to the dashboard and patient management modules, as those are the tasks typically performed by a receptionist in the system. All these roles and rights are assigned by the administrator to ensure that access is tailored to the intended user.

Receptionist will normally just register patients to the system if the patient is coming to the hospital for the first time but if it is not their first time then there is no need to register them all they need to do check the patient in to the hospital so that in the system everyone (nurses, doctors, etc) know that the patient has arrived at the hospital, once the patient has been given all the treatment they need and are leaving then they need to be checked out by the receptionist so that everyone in the system knows that the patient all of this can be done in the patient management modules as seen in Figure 2.2.

For a normally outpatient they will be checked in the hospital and then move onto triaging and the nurses who take vitals will be able to access the triage module of the system and in this module they will be able

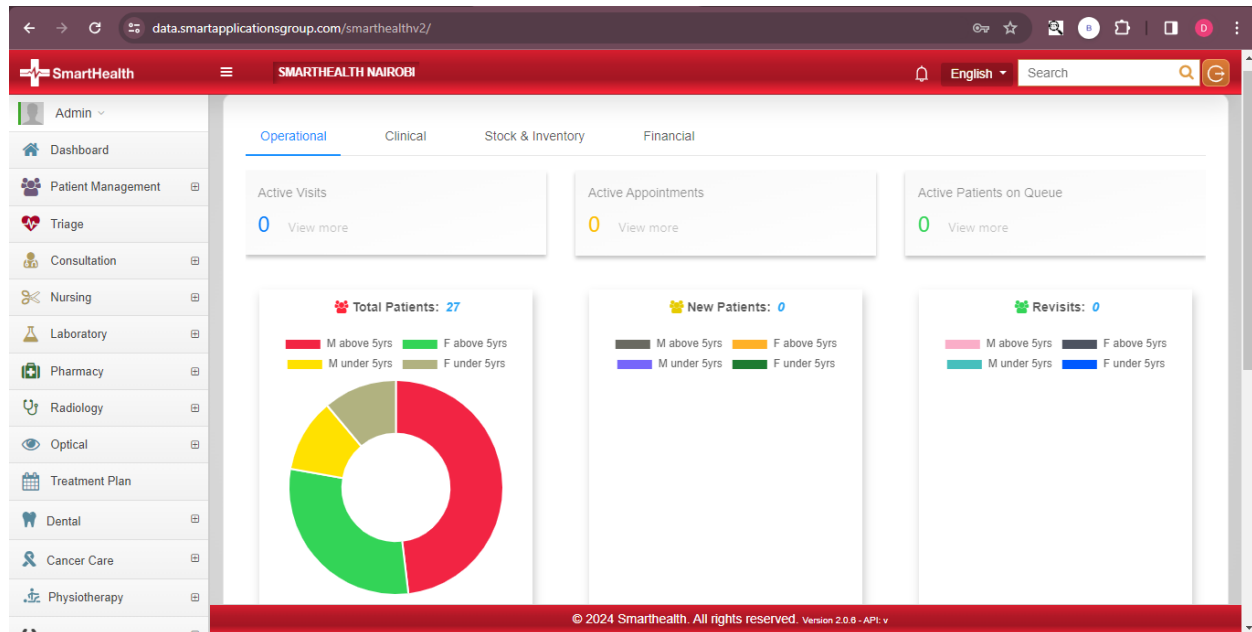


Figure 2.2

System Link

This system, known as "System Link" is a billing system for health insurers used by healthcare providers to determine a patient's active status with their health insurer, the benefits accessible under their subscribed health insurance plan, any applicable shortfalls or co-pays, and ultimately, to bill the patient for services rendered, eliminating the need for invoicing. In Figure 3.1, the landing page of the Smart Link system is depicted. Within the system, a patient can present their insurance card, placing it on an RFID card reader, and then scan their fingerprint. This process ensures that the patient using the card is the registered cardholder, thus mitigating any potential fraud. Alternatively, as shown in Figure 3.2, patients can be searched by their member access code, obtainable through the Smart Access app on their phone or by dialling the Smart USSD on their phone. In situations where a patient is sick but

does not have their card, this alternative method allows for fingerprint scanning to verify identity. The final method of patient search, as depicted in the drop-down menu in Figure 3.3, is utilized as a last resort if the other two methods fail to yield results.

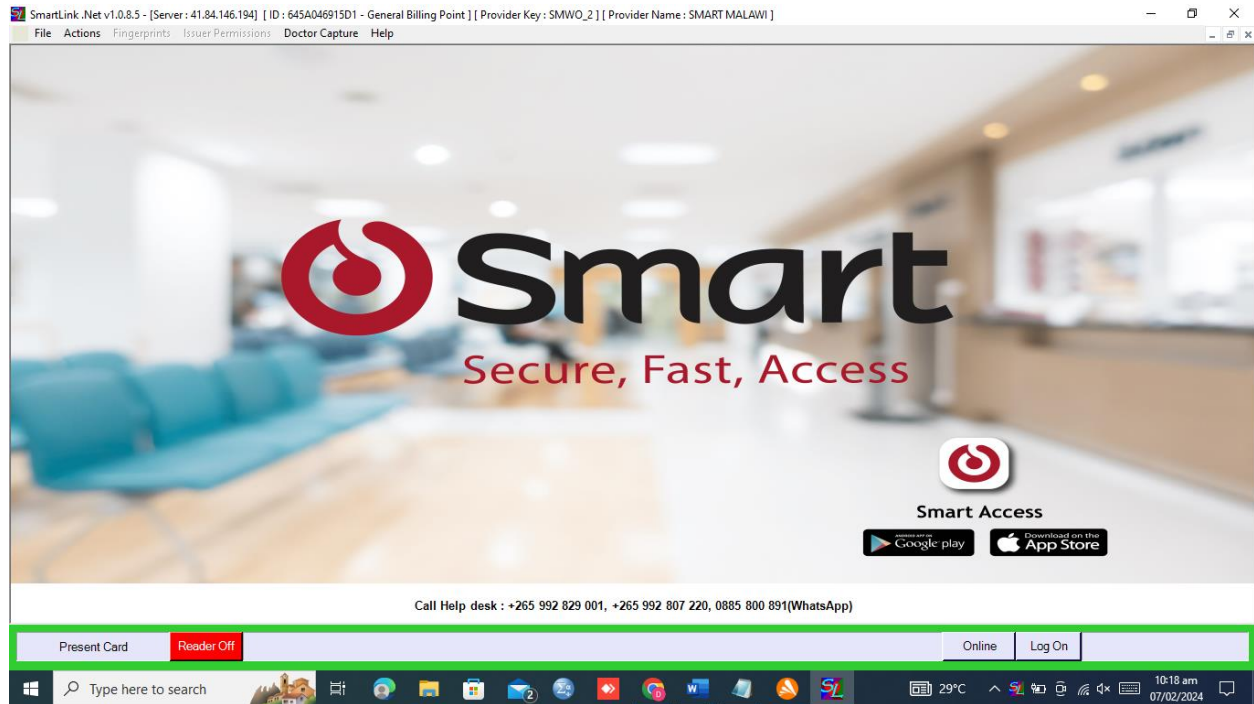


Figure 3.1

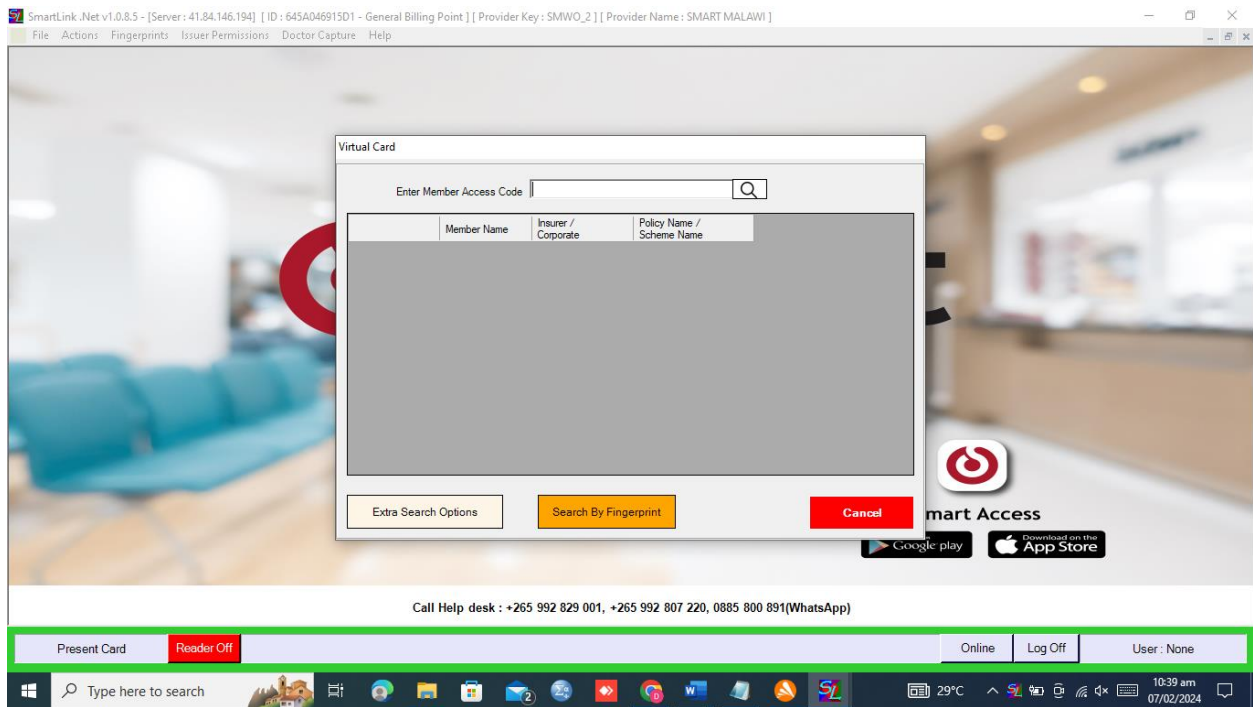


Figure 3.2

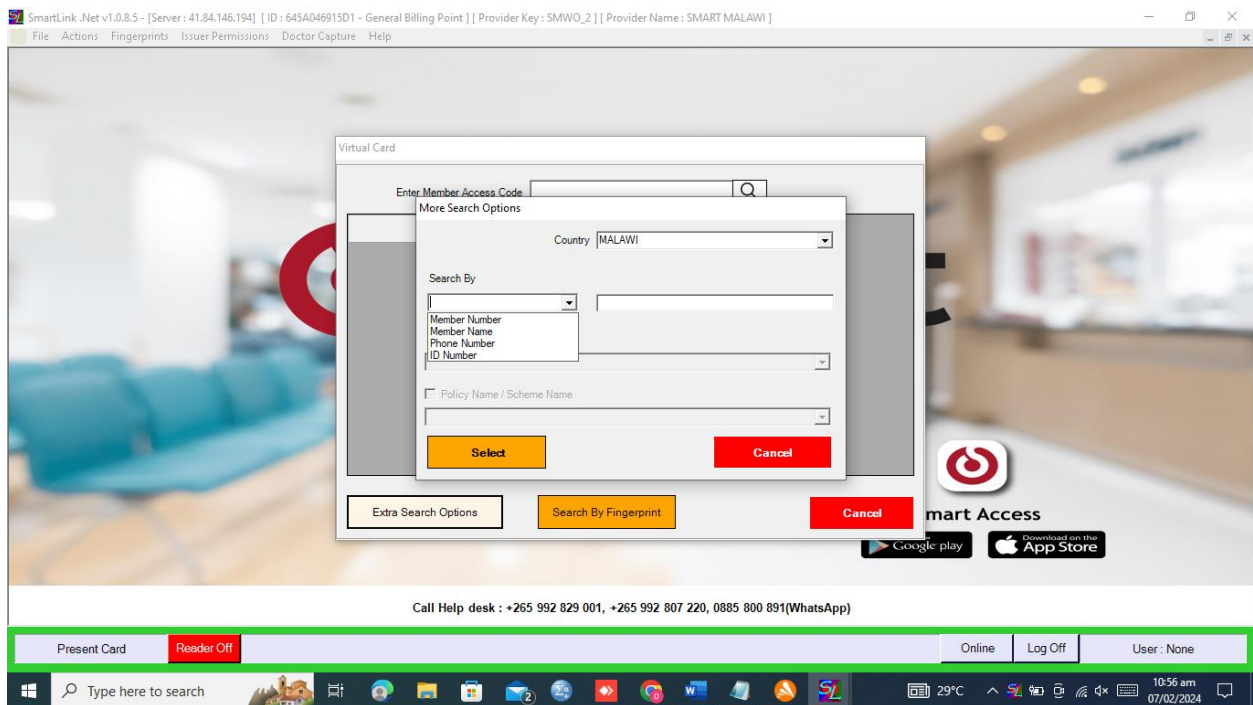


Figure 3.3

In Figure 3.4, I conducted a test to verify the patient's active status. If the patient is indeed active, the system redirects us to a page where we can view their benefits under the tab labelled "Claimable Benefits." On the right-hand side, you will find the "Capture Claim" button alongside the patient's information. Additionally, at the bottom in red, we can identify the benefits that the patient cannot claim under their type of insurance. At the reception the receptionist I will select the benefits that can be accessed by the patient for example "outpatient overall" then once the benefits are selected then the receptionist can capture the claim of all the services the were used by the patient in the that particular visit and can send it to the insurer.

SmartLink .Net v1.0.8.5 - [Server: 41.84.146.194] [ID : 645A046915D1 - General Billing Point] [Provider Key : SMWO_2] [Provider Name : SMART MALAWI]

File Actions Fingerprints Issuer Permissions Doctor Capture Help

Rules / Exclusions Back to Start

Claimable Benefits : SAMW-XMW.SAMPLEM

Nr	Benefit Description	Amount (Malawi Kwacha)	Link	Statutory Cover
1	OUTPATIENT OVERALL	70,000,000.00	56	No
2	OUTPATIENT DENTISTRY	600,000.00	1	No
3	OUTPATIENT OPHTHALMOLOGY PROCEDURES	1,500,000.00	1	No
4	OUTPATIENT SPECTACLES	200,000.00	1	No
5	OUTPATIENT CHRONIC TREATMENT	1,000,000.00	1	No
8	CT SCAN (CTS)	750,000.00	1	No
9	MRI SCAN (MRI)	1,000,000.00	1	No
10	PSYCHIATRIC AND MENTAL HEALTH	6,000,000.00	1	No
12	COVID 19 COVER	50,000,000.00	56	No
13	OUTPATIENT HEARING AIDS	600,000.00	1	No
14	OUTPATIENT PHYSIOTHERAPY	1,000,000.00	1	No
15	MSP WALLET	10,000.00	No	No
16	OUTPATIENT ORTHOPAEDICS	4,000,000.00	1	No
17	ORTHOPAEDICS APPLIANCES	300,000.00	1	No

View Only Benefits Benefits

Nr	Benefit Description	Amount (Malawi Kwacha)	Link	Statutory Cover
18	HOSPITALIZATION	50,000,000.00	56	No
24	INPATIENT OPHTHALMOLOGY PROCEDURES	1,500,000.00	18	No
25	INPATIENT SPECIALIZED DENTISTRY	50,000,000.00	18	No
31	INPATIENT PHYSIOTHERAPY	1,000,000.00	14	No
32	INPATIENT ORTHOPAEDICS	4,000,000.00	16	No
33	PSYCHIATRIC AND MENTAL HEALTH	50,000,000.00	18	No
34	CT SCAN (CTS)	750,000.00	18	No
35	MRI SCAN (MRI)	1,000,000.00	18	No
36	INPATIENT ONCOLOGY/CANCER TREATMENT	15,000,000.00	18	No

Capture Claim

Gender: Male

Membership Number: SAMPLE0024-01

Number:

Member Name: DENNIS

Member Surname: SMART

ID Number:

DOB: 1998-05-06

Next of Kin Name:

Next of Kin Tel. Contact: Update

Member Phone Number: *****883 Update

Scheme Name: MALAWI SAMPLE SCHEME

Scheme Plan/Option: SAMW-XMW.SAMPLEM

Patient File Number: 09 Update

Card Serial Number: VCMW0000002958

RET: 0 MASS: 0

Last Visits

Provider	Date	Amount

Figure 3.4

Challenges Faced and Solutions

In the course of my internship, I encountered several challenges that demanded solutions

Computer Literacy Challenge

During my internship, a notable challenge arose during user training sessions for the smarthhealth hospital information management system, as a considerable portion of the users lacked the required computer literacy. In response, I adjusted my training approach by simplifying complex concepts, incorporating hands-on demonstrations, and fostering an environment conducive to questions and discussions.

Team work Challenge

a struggle with working effectively within a team dynamic. Recognizing the importance of collaboration and cohesion, I took immediate action to adapt my approach and align with the team-oriented culture at Smart Applications. This involved a deliberate effort to enhance communication, actively listen to team members' perspectives, and seek constructive feedback. I worked on fostering a more collaborative mindset, embracing the diversity of ideas within the team.

On-site visit issue

One notable challenge I encountered during my on-site visits to hospitals was the limited availability of hospital employees, often occupied with patient care responsibilities. Recognizing the need for a more strategic approach, I quickly learned to proactively call ahead of my visits to coordinate schedules and identify times when employees would be less occupied. This adjustment allowed for more efficient use of everyone's time and ensured that my presence wouldn't disrupt critical patient care activities

Application of theoretical knowledge in practice

In my first year, during the course titled "Communication," we delved into various aspects of business communication, with a focus on writing professional letters. I applied this theoretical knowledge when seeking internships, effectively crafting application letters that ultimately led

to securing an interview and internship at Smart Applications. Additionally, the course covered essential skills for job interviews, including preparation, presentability, and interview conduct. These skills proved invaluable during my successful in-person interview with Smart Applications International Ltd. However, despite applying these skills, an online job interview with D-Tree International was unsuccessful, as depicted in Figure 4.1. Moreover, the course also covered meeting etiquette, which I applied during weekly meetings at Smart Applications International Ltd., held every Wednesday.

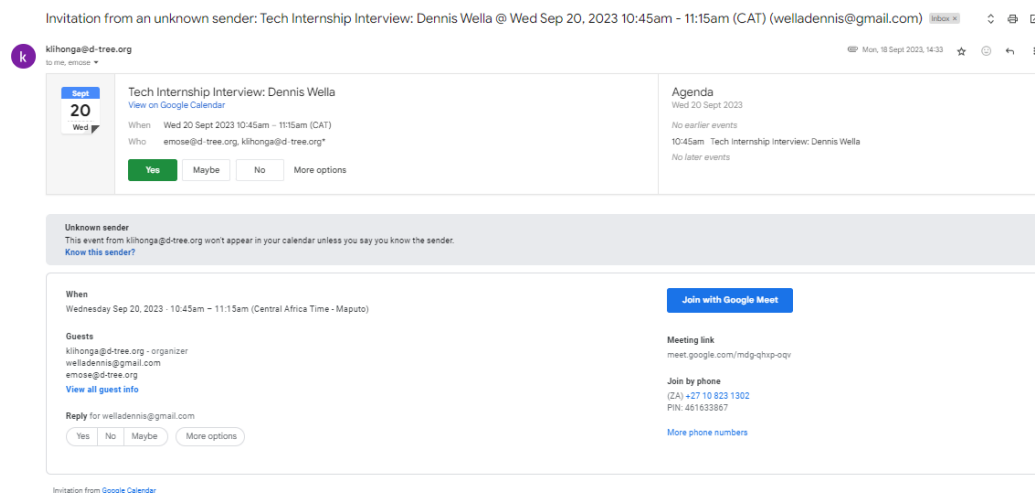


Figure 4.1

In my first year, during the course titled "Computer Systems Installation & Configuration," I acquired foundational knowledge in various concepts related to computer hardware, operating systems, software, basic networking, security, operational procedures, and customer service. I effectively applied this knowledge when tasked with setting up the Smarthealth and Smartlink systems. This involved leveraging my understanding of computer installation, operating systems, and maintenance & troubleshooting to ensure the successful implementation and ongoing functionality of these systems.

In my second year, during the course titled "Organizational Behaviour," I gained valuable theoretical knowledge that I successfully applied during my internship at Smart Applications. This included concepts such as group dynamics, leadership, communication, conflict resolution, and organizational resources. Utilizing this knowledge, I effectively navigated team dynamics, demonstrated leadership skills when required, communicated effectively with colleagues and clients, resolved conflicts as they arose, and optimized organizational resources to enhance productivity and efficiency within the workplace.

In my second year, during the course titled "Databases," we delved into the principles of data management in a database environment. This encompassed analyzing data requirements of an information system, developing an appropriate database schema to meet these requirements, and database application development. I directly applied some of these theoretical concepts during my internship at Smart Applications, particularly when implementing Smarthealth systems. Utilizing the HeidiSQL open-source database, as depicted in Figure 4.2 and 4.3, I effectively implemented these principles to ensure the successful development and deployment of the Smarthealth systems.

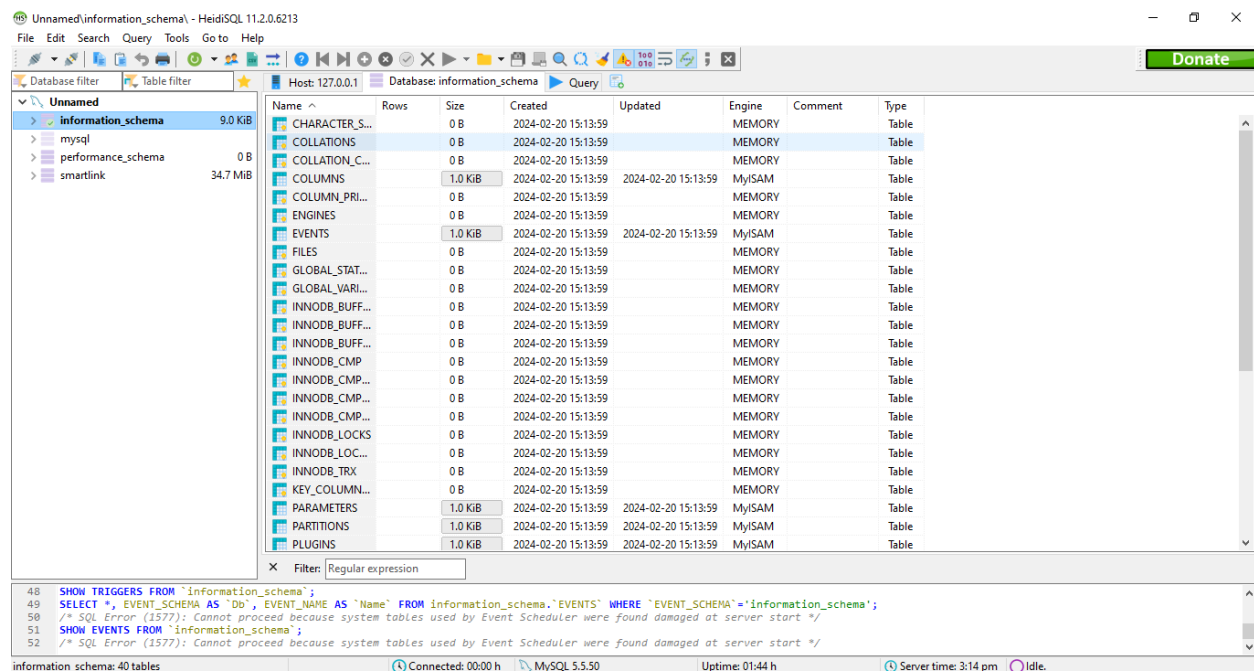


Figure 4.2

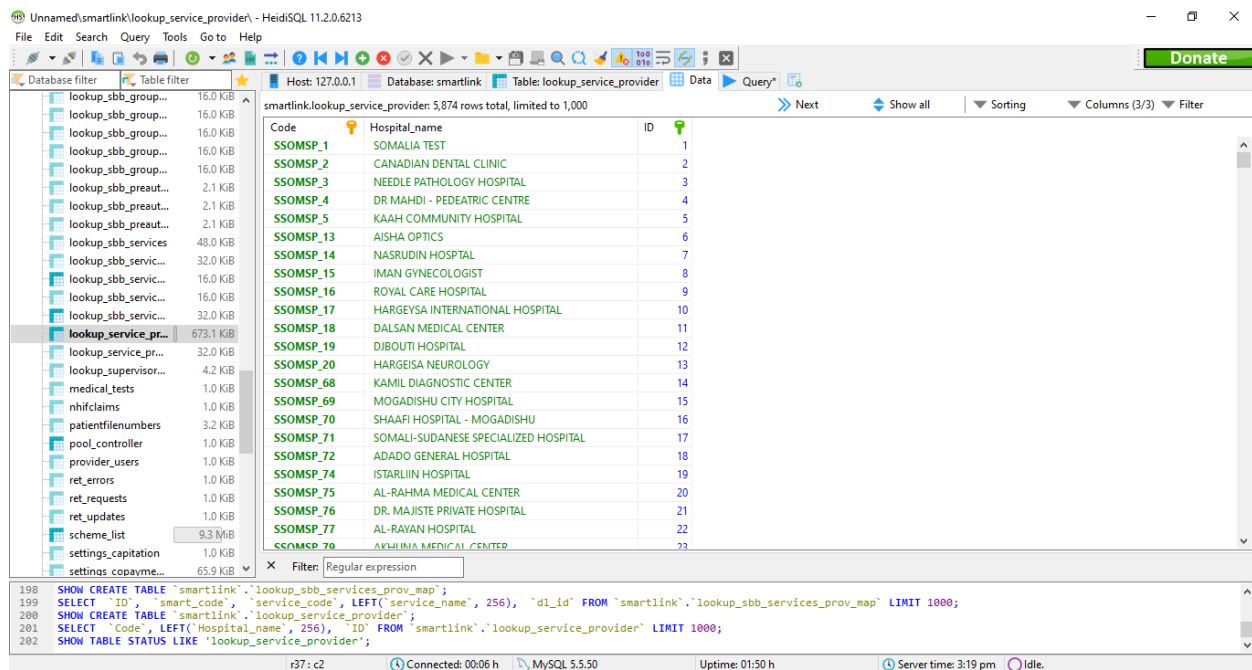


Figure 4.3

In the third year, during courses like "Systems Project Management," "Routing & Switching Network Analysis," "Computer Security & ICT Ethics," and "Enterprise Server Management & Network Security," I gained theoretical knowledge that proved beneficial during my internship. In Systems Project Management, I applied concepts practically during the Smarthealth system project presentation, where I effectively presented the system to all users and provided training on its usage. In Routing & Switching Network Analysis, where we studied routing and switching technologies, I utilized these concepts in implementing the Smarthealth system, which operates on a local area network post-installation. Additionally, in Enterprise Server Management & Network Security, I acquired knowledge and experience in working with enterprise-level computer systems, including workstation and file server management, web servers, networking devices, configuration management, monitoring, and more. This comprehensive understanding enabled me to contribute effectively to the deployment and maintenance of systems during my internship.

Conclusion

Summary of Experience

During my internship, I played a pivotal role as a 24/7 technical support resource for the smartlink system and smarthealth hospital information management system. This involved addressing and resolving technical issues over the phone, ensuring seamless system functionality. Additionally, I conducted on-site visits to hospitals, performing necessary system updates and troubleshooting challenges. I proactively identified hospitals lacking a hospital information management system and actively contributed to its implementation. This encompassed training users, setting up local area networks, assisting with server setup, and entering crucial data. Throughout the internship, I successfully navigated challenges, such as low computer literacy among users, adapting my training methods accordingly. My experience emphasized the importance of adaptability, effective communication, and strategic problem-solving in contributing to the successful integration and maintenance of healthcare information management systems.

Lessons Learned

My internship provided invaluable lessons that significantly contributed to my professional growth. One key takeaway was the importance of adaptability. Navigating diverse challenges, from technical issues to user training dynamics, taught me to be flexible in my approaches, adjusting strategies based on specific needs and contexts. I learned the critical role of effective communication, especially in a healthcare setting, where clear and concise information is paramount. Engaging with users, healthcare professionals, and colleagues honed my interpersonal skills and emphasized the significance of collaboration. The experience also underscored the value of proactive problem-solving; facing challenges head-on and finding innovative solutions became second nature. Moreover, the internship enhanced my understanding of the healthcare industry, emphasizing the need for a holistic approach that considers both technological and human elements. Overall, these lessons have not only

enriched my technical skill set but have also shaped my approach to teamwork, communication, and adaptability in a professional setting.

Impact on Work Readiness/Employability

This internship significantly boosted my work readiness and employability by honing my technical, problem-solving, and project management skills. Serving as a 24/7 technical support resource improved my troubleshooting abilities, while on-site visits and system implementations enhanced my adaptability and project management capabilities. Proactively identifying business opportunities and addressing user challenges showcased my communication and problem-solving skills. Dealing with low computer literacy among users improved my training and communication abilities. Overall, this experience has not only enriched my technical proficiency but also cultivated adaptability, effective communication, and problem-solving skills crucial for success in the professional realm.

Recommendation

Reflecting on this immersive experience, I recommend that future industrial attachment programs prioritize assisting students in finding their internships or providing guidance on how to secure one. The internship program coordinator should play a proactive role by distributing logbooks before the internship begins, facilitating better preparation and understanding of expectations. This proactive approach would empower students, ensuring a smoother transition into their internships and maximizing the overall learning experience. Providing guidance on securing internships and early access to logbooks would be invaluable in enhancing the effectiveness and success of future industrial attachment programs.

In conclusion, this portfolio encapsulates the highlights and accomplishments of my academic and professional journey. Through various experiences, projects, and challenges, it reflects my growth, skills, and commitment to continuous learning. I am excited about the future, equipped with a diverse skill set and a passion for making impactful contributions in my chosen field. This portfolio is a testament to my dedication, adaptability, and the valuable lessons learned along the way.

References

- Smart Applications International Ltd. (2023). Retrieved from <https://smartapplicationsgroup.com/>

Appendices